

Secure and Private Benchmarking for Subscription-Based AI Inference Services

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The AI Inference Service Market

- AI models are increasingly accessed via **subscription-based APIs**
- A typical workflow:
 - i. Client evaluates model quality during a **free trial**
 - ii. Client subscribes and **pays** for continued access

A Two-Sided Vulnerability

Malicious Server can:

- Use a strong model during trial, swap to a weaker one after payment
- Replay cached answers if client repeats queries

Malicious Client can:

- Extract needed answers during evaluation, then claim quality is too low
- Fabricate complaints to get refunds

Problem Statement

Goal: Design a protocol that simultaneously guarantees:

- The client cannot extract useful answers without paying
- The server cannot (i.e. it will be caught) degrade quality after payment
- Efficient mechanism (current solutions are too complex and inefficient)

Key Insight

Separate the query set into **pre-purchase** and **post-purchase** subsets via a **public coin flip**, committed before any model output is seen.

A smart contract on a blockchain acts as a **neutral, incorruptible arbiter**.

Protocol Phases

Phase 1

Commitment & Challenge Reservation

- Merkle tree of N query pairs
 - Commit benchmark B, τ
 - Lock deposits
- Coin-flip $\rightarrow$ split queries pre / post



Phase 2

Pre-Purchase Evaluation (2PC)

- Reveal pre-purchase queries
 - Server runs model M
 - 2PC checks $B(\text{Ans}) \geq \tau$
 - Output: single bit b
- If $b=1$: subscribe + pay



Phase 3

Post-Purchase Accountability

- Use post-purchase queries
- Server returns signed answers
 - Client builds ZK-SNARK
 - Smart contract arbitrates
 - Refund + slash if fraud

Phase 1 — Commitment & Challenge Reservation

- Client constructs **N pairs** of queries: $(q_i^{(0)}, q_i^{(1)})_{i=1}^N$
- Builds a **Merkle tree** over all $2N$ leaves and publishes root R_Q
- **Fair coin-toss** $\rightarrow$ public bit string $r \in \{0, 1\}^N$
 - which query is **pre-purchase** ($\{q_i^{\text{pre}}\}_{i=1}^N$.) vs **post-purchase** ($\{q_i^{\text{post}}\}_{i=1}^N$.)
- Neither party can bias the split
- Commits to benchmark $C_B = \text{Com}(r; r_r)$ and threshold $C_\tau = \text{Com}(\tau; r_\tau)$
- Both parties **lock deposits** into a Smart Contract S

Phase 2 — Private Pre-Purchase Evaluation (2PC)

1. Client reveals $\{q_i^{\text{pre}}\}_{i=1}^N$ with Merkle proofs (proves queries belong to committed set)
2. Server runs model $M \rightarrow$ produces Ans^{pre} , submits signed commitment σ_1
3. **Two-Party Computation** evaluates $B(\text{Ans}^{\text{pre}}) \geq \tau$ with private inputs on both sides

✓ Output: a single bit b

If $b = 1$: joint attestation α submitted on-chain $\rightarrow$ client subscribes and pays.

If $b = 0$: no attestation produced, no payment.

Only the **decision bit b** is revealed — no model outputs, no benchmark details.

Phase 3 — Post-Purchase Accountability

If the client suspects model degradation after paying:

1. Client sends post-purchase queries $\{q_i^{post}\}$ to server
2. Server returns **signed answers** $\sigma_i = \text{Sign}_{sk_{Ser}}(H(q_i^{post}), a_i)$
3. Client constructs a **ZK-SNARK proof** $\pi_{complaint}$ certifying:
 - Correct openings of C_B, C_τ , Merkle proofs
 - All server signatures are valid
 - $B(\{q_i^{post}\}, \{a_i\}) < \tau$

Smart Contract Arbitration

If $\pi_{complaint}$ verifies → **refund client + slash server deposit**

Building Blocks

Merkle Tree

Commits to N query pairs
Proof size: $O(\log N)$
Binding: no fake queries
SHA-256 hash function

Commitments

Hides B, τ from server
Binding: can't switch
benchmark after commit
Hiding + Binding

2PC

Evaluates benchmark
Output: single bit b
Client learns nothing
about model answers

ZK-SNARK

Proves degradation
without leaking B or τ
Succinct on-chain
verification (Groth16)

Smart Contract (Blockchain)

Stores commitments · holds deposits · verifies attestations and ZK proofs · issues refunds · slashes collateral

Scurity Properties



Model Privacy

Weights, architecture & activations never disclosed.
2PC security guarantees this.



Output Privacy

Client learns only bit b before payment. No free answer extraction.



Benchmark Privacy

Server learns only bit b .
 B and τ stay hidden via commitments + ZK.



Query Binding

Client can't inject queries outside committed set.
Merkle binding enforces.



Non-Repudiation

Server signs every answer.
Cannot deny outputs used in a complaint.



Accountability

Degradation detected with
 $\text{prob.} \geq 1 - 2^{-128}$
($N=5000$, $\epsilon'=0.1$)

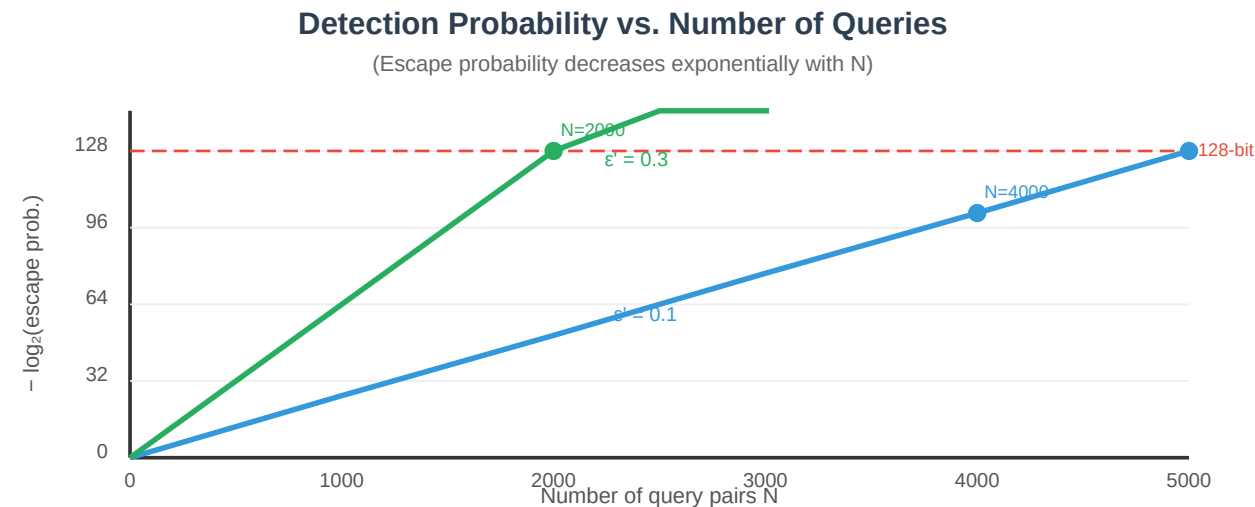
Detection Probability: Chernoff Bound Analysis

Let $\varepsilon' = \tau - (1 - \varepsilon)$ be the **degradation gap**.

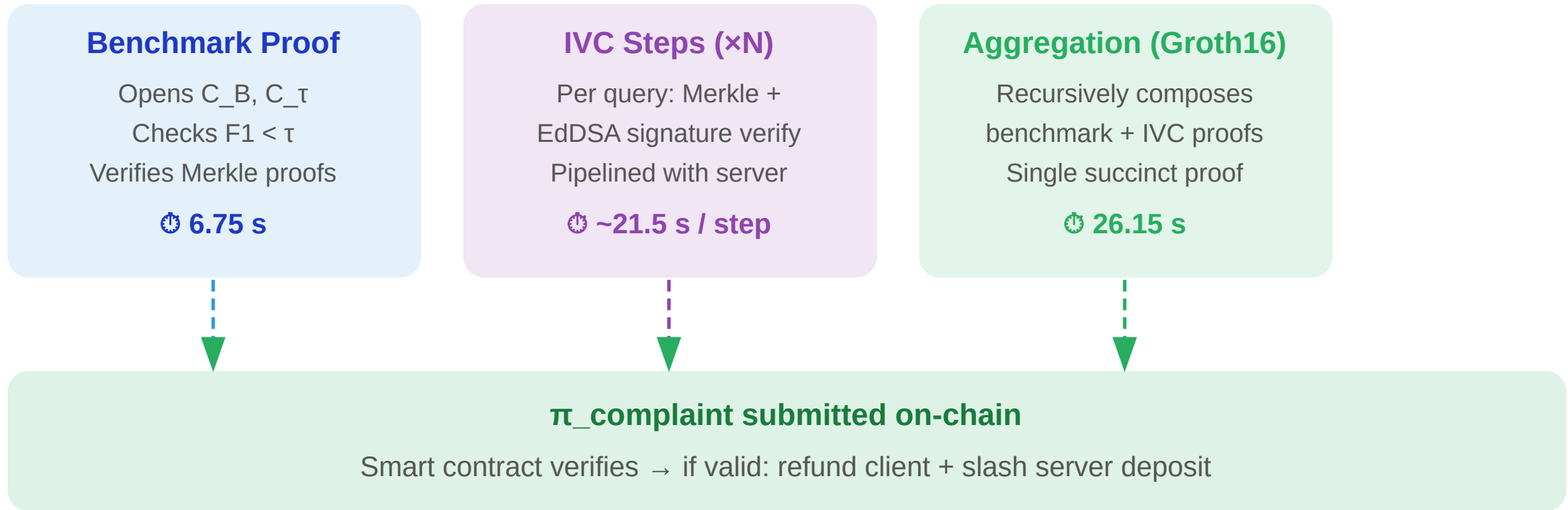
By the Chernoff bound:

$$\Pr[\text{post-purchase score} \geq \tau] \leq \exp\left(-\frac{N \varepsilon'^2}{2\mu + \varepsilon'}\right)$$

The **larger the degradation**, the **fewer queries** needed to catch it.



ZK SNARKS compliant mechanism



Total ≈ 7.7 min (20 queries)
Peak RAM: 8.6 GB

ASAI's features:

- Does **not** arithmetize the entire neural network
- Provides **pre-purchase** quality assurance (novel)
- Combines 2PC + Merkle + ZK-SNARK + Smart Contract
- Lightweight: ZK proof generated **only if there is a complaint**
- Fully **model-agnostic** (black-box access to M)

Comparison with Related Work

Approach	Technique	Model Privacy	Proving Cost	Goal
zkCNN, zkLLM, zkGPT	Full ZK-SNARK per inference	✓ Weights hidden ✗ Circuit = full model	Minutes / inference (GPU)	Verify outputs
opML	Optimistic fraud proofs	~ Partial	Low (dispute only)	On-chain ML
Lightweight ZK (ACC+26)	Sampled Merkle paths	✓ Weaker soundness	Milliseconds	Verify outputs
PrivaDE (WTCS26)	2PC + ZK + blockchain	✓ (data, not model)	Moderate	Data markets
ASAI (ours)	2PC + Merkle + ZK-SNARK + Smart Contract	✓ Full (no weights/arch)	~7.7 min (complaint only)	AI service accountability

Summary & Contributions

- **ASAI protocol:** first construction providing both pre- and post-purchase accountability for AI inference services
- **Benchmark and output privacy:** only a single bit disclosed before payment
- **On-chain arbitration:** smart contract enforces refunds and collateral slashing without trust in any party
- **Practical viability:** ZK complaint proof in ~7.7 min, fully pipelined, on commodity hardware

Future Directions

- **Benchmark:** how to define a reliable benchmark $(q_i^{(0)}, q_i^{(1)})_{i=1}^N$?
- **Reduce 2PC overhead:** explore MPC-in-the-head or VOLE-based protocols for the benchmark evaluation step
- **Formal verification:** machine-checked proofs of the protocol security properties
- **Regulatory compliance:** map ASAI guarantees to emerging AI Act audit requirements

Thank You!

Questions?

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 <https://github.com/marcozecchini/secureVerificationModel>